



Code A Pulse Oximeter

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What is a Pulse Ox?

A Pulse Oximeter is a device that measures a person's heart rate (Beats Per Minute) and how much Oxygen is in their blood (SpO₂).

How do you think this device measures your pulse with LIGHT?





Variables & Data Types

What is a variable?

A **variable** is either one letter or multiple letters

- A variable can be anything from "x" to "temperature"
- Serve as a "bucket" for holding information

What is a data type?

A **data type** specifies how the data is stored in a variable



Data Type "int"

—

"int" can be thought of as short for “integer”

A **whole number** (no decimals or fractions)

Similar to:

- An int is like a jar that can hold only whole candies, nothing cut in half

Examples: 12, 0, -5

When we use it:

Pin numbers on a circuit (`int RED_LED = 12;`), counting things, storing whole-number



Data Type "float"

—

"float" is short for “floating-point number”

A number with **decimals**, more precise values

Similar to:

- A float is like a money, that lets you have things like cents, a nickel = 0.05 dollars

Examples: 3.14, 0.75, 98.6

When we use it:

Calculations that need decimals, measurements (temperature, oxygen levels, averages), anything that isn't a whole number



Data Type "bool"

—

"bool" is short for "boolean"

Only true or false

Similar to:

- A bool is like a light switch, it can only be ON or OFF, nothing in between

Examples: `bool isLightOn = true;`

When we use it:

Checking if something is ON or OFF, storing yes/no decisions



Activity

Task: identify which data type is appropriate

Example 1 — LED Pin Number:

```
LED_PIN = 9;
```

Example 2 — Sensor Reading With Decimals:

```
temperature = 23.7;
```

Example 3 — LED On or Off:

```
buttonPressed = false;
```



Activity Answers

Example 1: (LED_PIN = 9;)

- Correct answer = **int**
- Why? → LED pins are whole numbers → integers

Example 2: (temperature = 23.7;)

- Correct answer = **float**
- Why? → Temperatures often use decimal values → floating point

Example 3: (buttonPressed = false;)

- Correct answer = **bool**
- Why? → A boolean stores true or false



Digital Outputs

Digital Outputs (What They Are):

- A digital output sends either HIGH or LOW signals
- HIGH = ON (5V)
- LOW = OFF (0V)

How This Relates to LEDs:

- We connect an LED to a digital pin on an Arduino
- Arduino sends a HIGH signal → LED lights up
- Arduino sends a LOW signal → LED turns off

Key Idea: Digital outputs let the Arduino **control** things



Demo: Turning an LED On/Off

Turning an LED ON:

```
digitalWrite(RED_LED, HIGH);
```

- This sends power through the pin → LED lights up

Turning an LED OFF:

```
digitalWrite(RED_LED, LOW);
```

- This stops the power → LED turns off



Analog Inputs

Analog Inputs (What They Are):

- Analog inputs read a range of values, not just ON/OFF
- Values go from 0 to 1023 depending on light intensity

How This Relates to the Light Sensor:

- The photodiode (light sensor) changes voltage based on how much light hits it
- Arduino uses `analogRead(A0)` to measure this

Real-World Meaning:

- More light → bigger number
- Less light → smaller number

Key Idea: Analog inputs let the Arduino sense the environment



Demo Analog Inputs and Print

Reading the Sensor:

```
int reading = analogRead(A0);
```

- Reads the light levels from the photodiode
- Stores the number in a variable

Printing the Value:

```
Serial.println(reading);
```

- Sends the number to the Serial Monitor
- Students see real numbers changing in real time



Guided Worksheet

Examples to fill in: (What were your answers and why?)

1. _____ RED_LED = 12; // choose correct data type
2. pinMode(RED_LED, _____); // input or output?
3. digitalWrite(RED_LED, _____); // turn LED on
4. int reading = analogRead(_____); // which pin?
5. Serial.println(_____); // print value
6. digitalWrite(RED_LED, _____); // turn it off



Test Code

—

What does the LED do as a result of the code?

- turn on/off

What is the Serial Monitor doing? What is this because of?

- Change on the Serial Monitor because of sensor readings

When placing a finger over the ALS (light sensor), what happens to the numbers?

- The numbers change



Implementation into PulseOx code

—

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- turn on/off

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